

ClariFi: Refinement Changelog

1. Problem Statement Update

The problem from a policy and socioeconomic angle framed the preliminary problem statement. While accurate, it read as an academic argument rather than a consumer-facing pain point, making it harder for the reader to personally feel the urgency of the problem.

The refined problem statement restructures the narrative around one central concept: Data Darkness, where the idea is that Malaysian consumers are actively operating without the data they need to make sound financial decisions. Rather than listing problems in isolation, the statement presents them as a connected cycle where each pain point directly causes the next, repeating every month unchanged.

2. Features Update

Feature	Preliminary Version	Refined Version
AI Fast Expense Capture	Voice or receipt photo capture, user reviews parsed result and confirms.	Unchanged. Core frictionless capture remains the foundation of the product.
Price Comparison Feature	- Converts captured bills and promotional brochures into price observations. - Single items lookup, where the user searches for one item, see prices at different stores	- Powered by direct partnerships with local groceries or supermarkets and crowdsourced receipts, building a living, real-time local price database. - Bucket Feature Implemented, users can now add multiple items into a list, and build a price matrix across all nearby stores to calculate the total basket cost at each location. It then weights results by proximity and recommends the single store where the user can buy everything at the most worthwhile combined price in one trip.
Expense Breakdown	View and filter recorded expenses.	Unchanged. Surfaces spending patterns and category-level leaks on a monthly basis.
Alerts & Notifications	Create threshold alerts and receive event notifications when conditions are met.	Unchanged. Proactively notifies users of price changes and budget thresholds.
AI Purchase Insights	<i>Not in preliminary version.</i>	AI analyses each purchase, flags overspending, directs users to cheaper

		alternatives via Community Price Intelligence, and explains price increases through real economy trends.
Family Profile	Share a family profile to track each other's spending and share promotions or deals.	Retained as a supporting feature. Enables shared household visibility and expense splitting.

3. Judges Feedbacks and Improvements

- **Feedback:** Stronger Technical Validation and Better AI Logic
Improvements: AI & Technology Stack

Layer	Technology Used	AI/ Logic	Output
Capture Intelligence	IOS speech recognition model (mobile), Gemini-3.1-flash-light (vision OCR), NestJS parsing pipeline	AI: Speech-to-text transcription + vision-based receipt reading. Logic: Named entity extraction pulls merchant, amount, date, category, and payment method from unstructured input. Low-confidence fields flagged for user review. Human-in-loop confirmation before saving ensures data integrity.	Structured, validated expense entry
Purchase Intelligence (AI Purchase Insights)	PostgreSQL price observation database, Gemini-3.1-flash-light (LLM reasoning), community receipt + grocery partnership data	AI: LLM generates a brief explanation of price deviations and economy-driven price trends. Logic: Price percentile comparison, where user's paid price is ranked against the distribution of the same item across nearby stores within a defined radius and recent time window. If price has risen broadly, time-series trend analysis identifies whether the increase is sustained or a one-time spike. Cheaper alternatives retrieved via geospatial query.	Spending verdict + cheaper alternatives + price trend explanation

Basket Intelligence (Price Intelligence)	PostgreSQL price matrix query, OSM Nominatim (geospatial), NestJS aggregation logic	Logic: Price matrix built across all key partners for every item in the list. Total basket cost calculated per store. Result weighted by proximity, where a further store must offer proportionally greater savings to rank higher. Stores ranked by best combined basket value and distance score.	Single store recommendation with full basket breakdown and savings comparison
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